

DOOR COUNTY CHERRYLAND AP, WI, USA

WMO: 726458

Lat: 44.850N

Lon: 87.417W

Elev: 724

StdP: 14.32

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04824

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.0	-0.4	-17.0	2.2	-4.2	-10.8	3.2	-0.5	26.3	18.7	24.4	20.8	10.0	260	0.605

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	84.2	71.7	81.7	69.8	79.3	68.0	74.2	81.9	72.1	79.3	70.1	76.6	11.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.0	121.4	80.1	69.9	112.9	77.4	67.7	104.6	74.7	38.1	82.0	36.3	79.5	34.5	77.4	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.5	21.2	19.2	DB	-10.8	90.3	6.6	3.5	-15.6	92.8	-19.5	94.8	-23.2	96.8	-28.0	99.3	
			WB	-11.1	78.0	5.2	2.0	-14.9	79.4	-17.9	80.5	-20.8	81.6	-24.6	83.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.1	19.1	21.3	30.2	41.8	53.4	63.6	69.0	67.6	61.2	49.4	36.9	25.8
	DBStd	19.46	11.35	11.10	10.09	8.00	7.74	6.88	5.64	5.28	7.21	7.98	8.70	9.96
	HDD50	3961	958	804	616	267	50	1	0	0	4	111	400	750
	HDD65	7655	1423	1224	1078	697	369	106	25	32	156	487	843	1215
	CDD50	2161	0	0	3	20	156	408	588	547	340	92	7	0
	CDD65	382	0	0	0	0	10	63	149	114	42	4	0	0
	CDH74	2177	0	0	0	2	83	403	968	534	176	11	0	0
	CDH80	461	0	0	0	0	12	83	243	91	31	1	0	0
Wind	WSAvg	9.6	10.3	10.3	10.0	10.5	9.5	8.3	7.9	7.6	8.9	10.2	11.0	10.5
Precipitation	PrecAvg	31.2	1.7	1.2	1.8	2.8	3.4	3.9	3.5	3.1	3.1	2.8	2.0	1.7
	PrecMax	37.7	4.0	2.1	4.1	5.1	6.1	8.9	6.6	5.9	6.7	6.5	5.4	3.8
	PrecMin	24.6	0.4	0.4	0.4	0.6	1.5	1.0	1.0	0.4	1.2	0.6	0.3	0.2
	PrecStd	3.1	1.0	0.5	0.9	1.1	1.3	2.3	1.6	1.2	1.3	1.7	1.2	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.0	47.6	60.8	70.4	81.4	86.4	90.3	87.6	84.2	74.6	63.2	52.1
		MCWB	40.3	42.9	52.2	57.2	67.3	73.2	76.2	71.8	71.1	65.0	53.6	49.6
	2%	DB	39.2	42.7	52.5	63.2	75.3	82.0	85.6	82.0	79.1	69.8	55.5	45.1
		MCWB	36.1	38.0	46.1	51.6	62.7	69.0	72.5	70.7	67.7	61.5	49.6	41.6
	5%	DB	36.5	38.2	47.3	57.5	71.7	79.1	82.1	79.5	75.0	64.3	52.2	41.2
		MCWB	34.4	35.1	42.0	47.5	60.8	67.2	70.4	68.6	65.9	57.7	47.5	38.1
	10%	DB	34.3	36.0	43.3	54.2	66.2	74.9	79.9	76.7	72.3	61.3	48.3	37.5
		MCWB	32.4	33.5	38.1	45.5	57.2	65.1	68.7	67.2	64.5	55.9	44.6	35.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.4	44.0	53.4	58.9	69.2	75.4	78.0	75.9	73.3	67.4	55.2	49.6
		MCDB	44.1	47.1	59.0	67.2	78.6	83.2	87.7	83.1	81.9	72.4	60.6	51.7
	2%	WB	37.1	38.4	46.7	53.5	65.2	71.3	74.8	73.2	69.9	62.6	51.2	42.3
		MCDB	39.1	41.6	50.7	60.6	73.3	79.3	82.6	79.7	75.9	67.3	55.3	44.6
	5%	WB	34.5	35.6	42.2	49.7	61.9	68.7	72.7	71.0	67.8	59.2	47.7	38.6
		MCDB	36.3	37.9	46.8	56.2	69.1	76.4	80.2	77.2	73.4	63.8	50.8	40.9
	10%	WB	32.3	33.1	38.6	46.2	58.7	66.4	70.5	69.1	65.9	56.3	45.4	35.9
		MCDB	34.2	35.5	42.9	53.2	65.9	73.5	77.4	74.5	70.9	60.9	48.1	38.2
Mean Daily Temperature Range		MDBR	11.8	12.6	13.7	15.0	16.7	16.3	16.7	14.8	14.9	12.4	10.5	10.3
		MCDBR	12.6	12.2	17.7	22.1	23.1	19.6	19.1	17.1	18.1	16.6	15.3	12.6
		MCWBR	11.3	10.1	13.0	14.5	14.6	11.7	10.5	9.5	10.5	11.7	12.0	11.3
	5% WB	MCDBR	11.9	11.2	16.6	21.3	21.6	17.9	17.6	15.0	15.7	15.6	13.9	11.7
		MCWBR	11.0	9.7	13.1	15.4	15.3	12.2	11.0	9.5	10.3	12.0	11.8	11.1
Clear-Sky Solar Irradiance	taub	0.270	0.293	0.325	0.367	0.395	0.403	0.393	0.393	0.370	0.332	0.303	0.271	
	taud	2.465	2.384	2.392	2.335	2.320	2.337	2.378	2.372	2.429	2.520	2.527	2.512	
	Ebn at Noon	273	285	289	283	278	275	276	272	270	268	259	262	
	Edh at Noon	22	29	33	38	40	39	37	36	31	25	21	19	
All-Sky Solar Radiation	RadAvg	447	759	1140	1438	1750	1897	1946	1656	1287	795	476	345	
	RadStd	50	81	104	136	141	142	81	109	100	80	46	36	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page